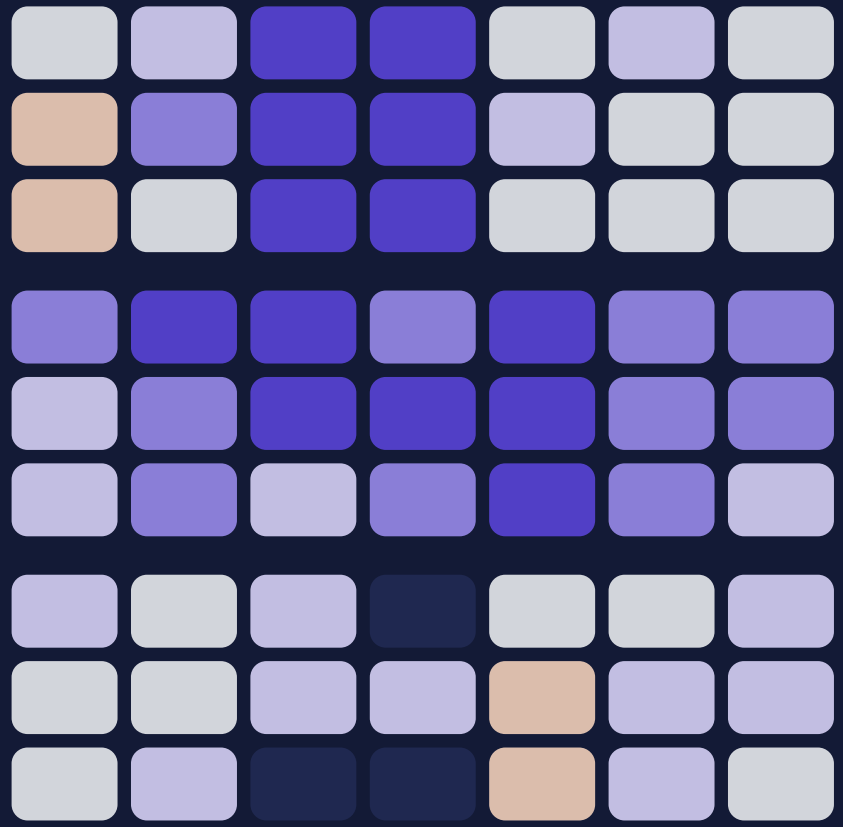
Texas MISA Fall 2026 Case Competition · Costco workforce planning challenge

Workforce Planner

A staffing plan for Costco's holiday week, Feb 12–18: every block in all three scenarios, set for service first, then cost.

Simulated data, not real Costco data



Warehouse 1, BASE: one cell per block, iris where the plan adds people, bronze where it cuts

The holiday week needs staffing that holds service through Valentine's Day and Presidents' Day

Presidents' Day window

Thu	Fri	Sat	Sun	Mon	Tue	Wed
Feb 12	Feb 13	Feb 14	Feb 15	Feb 16	Feb 17	Feb 18
		♡ Valentine's Day		⚐ Presidents' Day		

Three scenarios, 189 blocks each

BASE
Baseline planning week
Expected demand, with both holidays

PEAK
Promotion and winter-weather surge
Demand up to +30% on Feb 13–15, same staff

DISRUPT
Callouts and reduced availability
Fewer staff at Warehouses 2 and 4, Feb 12–16

Three departments, each with its own service target

Front End what members feel first
Checkout wait of 5 minutes or less

Merchandising keeps products on the shelves
Replenishment response of 60 minutes or less

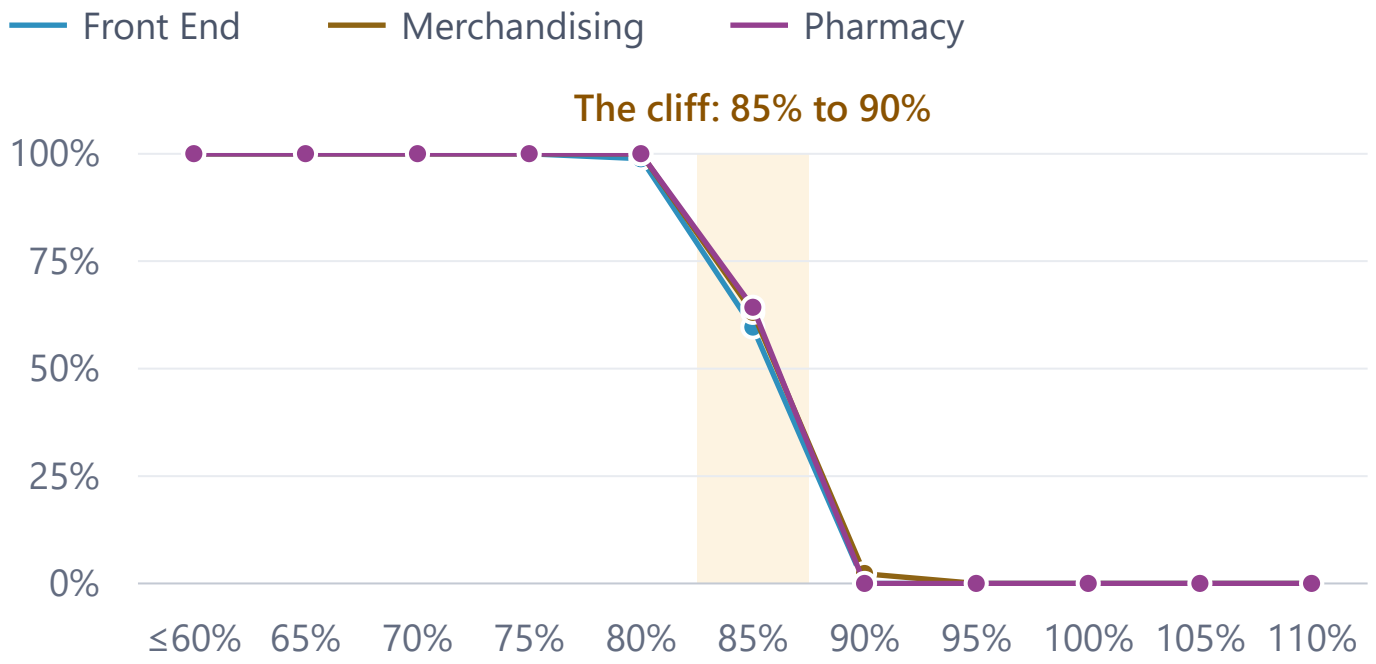
Pharmacy a health service members rely on
Prescription wait of 20 minutes or less

Simulated data, not real Costco data

Hours were about right but misplaced, and service collapses past 85% utilization

Share of blocks meeting target, by utilization

6 weeks of history. Each point is a 5-point band: 85% means 85 to 90%.



Hours above need, 6 weeks

3,184 h

$103K at regular rates

Hours short of need, elsewhere

2,472 h

⚠ 233 blocks missed target

Worked hours were within 1.2% of need: move hours before adding them.

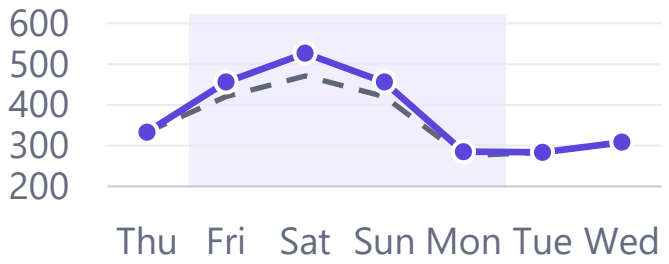
Simulated data, not real Costco data

Demand peaks on the holiday weekend, and the naive benchmark misses it

Our forecast Benchmark (case's naive forecast) Presidents' Day window

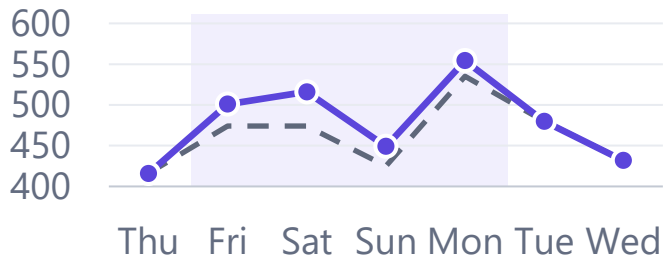
Front End

Checkout workload units per day



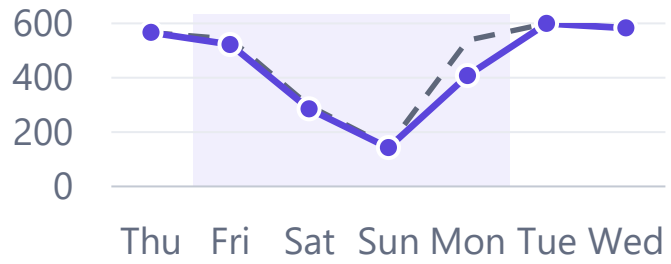
Merchandising

Replenishment workload units per day



Pharmacy

Prescription workload units per day



+12%

Front End, the weekend before Presidents' Day, measured on MLK weekend. Merchandising +8%.

+3%

Valentine's Day for Front End and Merchandising: a judgment call, with no history to measure it.

2.4% vs 4.1%

Daily error on held-out weeks, ours vs the benchmark. The gap is all Fri–Mon: holiday distortion in the benchmark.

Simulated data, not real Costco data

Our engine turns a forecast into a staffing plan that explains itself

Step 1

Forecast

Regression on warehouse, weekday, holidays and weekly level

Step 2

Service risk

Chance of meeting target, allowing for callouts and forecast error

Step 3

Cost rule

Regular staff, then spare cross-trained staff, then overtime

Step 4

Search

Tries all staffing pairs per shift and keeps the cheapest at 95%

Step 5

Explain

Every block gets its reason, staff sources, cost, risk and cushion

↗ Add 15 **Warehouse 2, Front End, Sun Feb 15, Midday · 32 in the draft to 47 in the plan**

The draft's 32 people would run at 108% utilization, with a 0% chance of meeting target. Adding 15 brings it to 73% utilization and a 97% chance.

Before a miss becomes the likelier outcome, this block can lose 7 people, or take 18% more demand.

Simulated data, not real Costco data

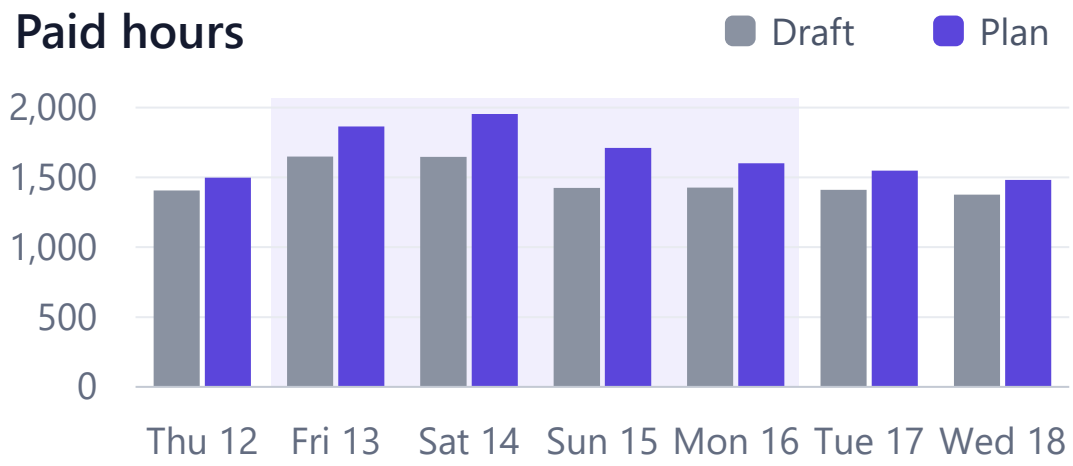
13% more hours, placed where they're missing, lift expected service from 68.5% to 95.6%

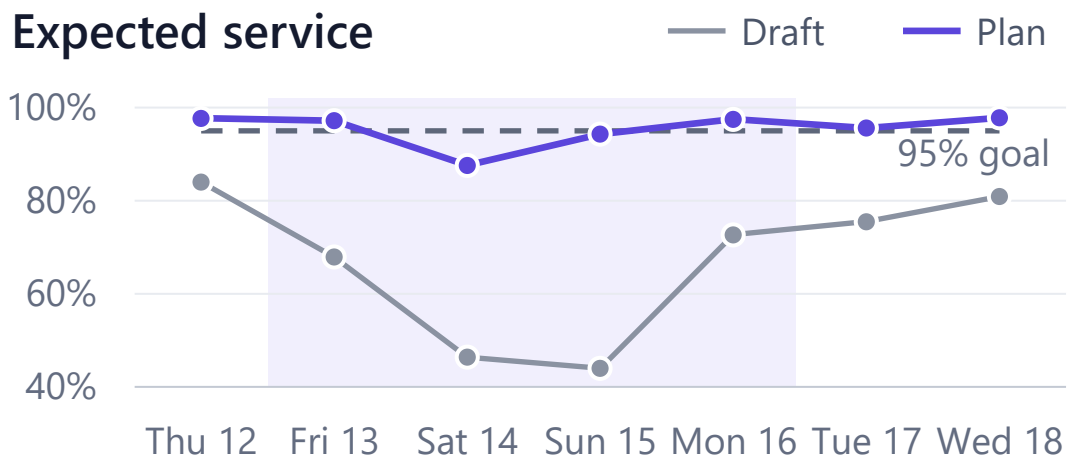
Labor cost, BASE
$377,504
↗ +$48.4K vs draft

Paid hours
11,658 h
↗ +1,314 h vs draft

Expected service
95.6%
↗ +27.1 pts vs draft

Blocks below the goal
25
↘ −110 vs draft





Simulated data, not real Costco data

The plan adapts to each scenario, and fields every person it schedules

BASE

The expected week

Expected demand with both holidays, as we forecast it

Expected service 68.5% to **95.6%**

	Draft	Plan
Labor cost	$329.1K	**$377.5K**
Paid hours	10,344	**11,658**
Blocks below goal	135	**25**
Over capacity	0	**0**

Hardest day: Sat Feb 14

PEAK

Promotion and storm

Demand up to +30% on Feb 13–15; same staff as BASE

Expected service 58.3% to **87.0%**

	Draft	Plan
Labor cost	$329.1K	**$387.9K**
Paid hours	10,344	**11,923**
Blocks below goal	139	**45**
Over capacity	0	**0**

Hardest day: Sat Feb 14

DISRUPT

Illness and short crews

Warehouse 2 short Feb 13–16, Warehouse 4 short Feb 12–13

Expected service 66.6% to **89.5%**

	Draft	Plan
Labor cost	$326.4K	**$366.6K**
Paid hours	10,207	**11,288**
Blocks below goal	142	**39**
Over capacity	12	**0**

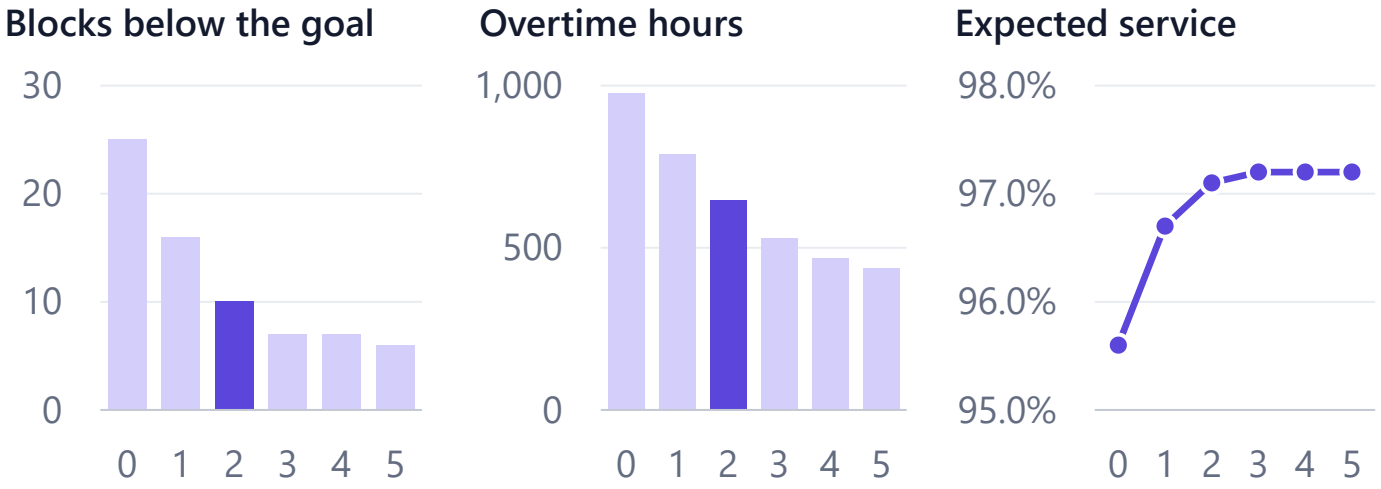
⚠ Draft can't field 36 people

Simulated data, not real Costco data

The best single action is limiting holiday PTO: service rises and cost falls

Keeping people from holiday PTO, BASE

The plan rerun with up to N people per block kept from holiday-weekend PTO



People kept from PTO per block, 0 to 5. Keeping 2 per block (darker bars): 10 blocks below the goal instead of 25, 647 h of overtime instead of 975, and $1.9K less for the week.

What a warehouse leader does

Before the week

- Keep 2 people per block off holiday-weekend PTO
- Approve the holiday weekend's overtime now, not on the day
- Line up outside cover for Sat Feb 14, where capacity runs out

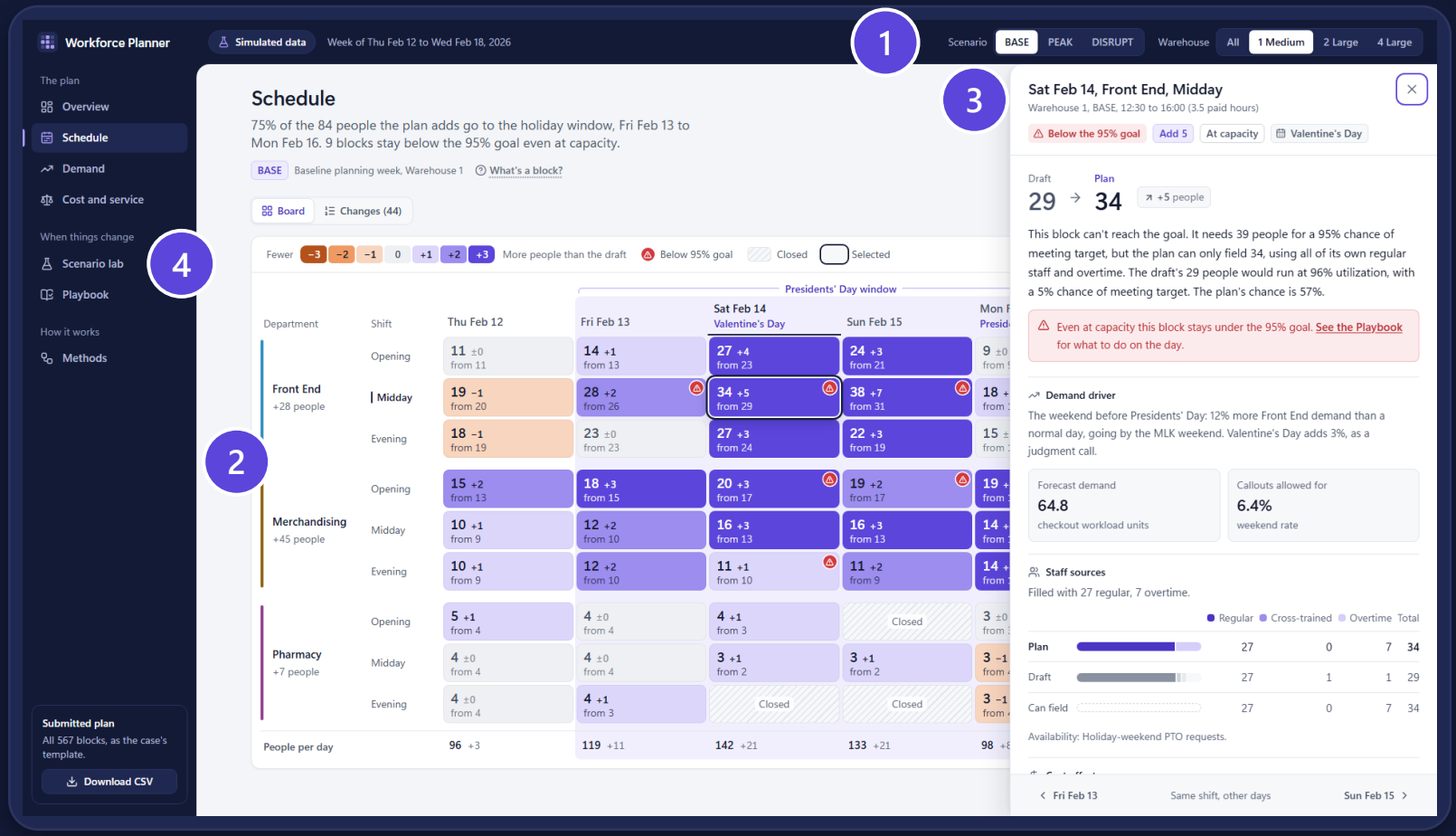
On the day

- Callouts past a block's cushion: cross-trained staff, then overtime
- Waits past target: pull help in now, not at the next shift
- Traffic well below forecast: offer time off in overtime blocks first

Simulated data, not real Costco data

The Workforce Planner: what you're about to see

1. **Scenario switch**
   BASE, PEAK or DISRUPT, for one warehouse or all
2. **The board**
   Every block of the week, colored by its change from the draft
3. **Why drawer**
   Each block's reason, staff sources, cost, risk and cushion
4. **Scenario lab**
   Change demand or callouts and rerun the plan live



Live demo, 5 minutes

Simulated data, not real Costco data

Service up 27.1 points for 14.7% more labor, and every person scheduled can work

+27.1 pts

Expected service, BASE: 68.5% in the draft, 95.6% in the plan

+14.7%

Labor cost, BASE: $329.1K in the draft, $377.5K in the plan

0

Blocks over capacity in any scenario. The draft has 12 in DISRUPT.

What we hand over

Staffing plan

All 567 blocks, three scenarios, in the case's template

Workforce Planner

The tool you just saw, running on one laptop

Report

Method, assumptions and results, for any reader

Notebook

The analysis code, run top to bottom

Simulated data, not real Costco data

Appendix

Supporting analysis for questions: the forecast method, the key assumptions, and cost and service against the draft. Each item on the right links to its slide.

A1 Forecast method

A2 Forecast accuracy and its limits

A3 Service curve and risk model

A4 Allocation and cost rule

A5 Key assumptions, ours marked

A6 Cost and service against the draft

A7 Where capacity runs out

A8 Service goal and callout allowance

A9 Validation and checks

A10 Data notes

A11 The submission file

The forecast is a regression on the calendar, with the holiday effect borrowed from MLK weekend

1 Fit

Log daily demand on warehouse, weekday, holiday window and week

2 Level

Predicted at the last 4 weeks' average level, best in the backtest

3 Holidays

From MLK weekend, plus Valentine's Day +3% for two departments

4 Shifts

Days split by average shift shares, with Front End holiday shares

Demand drivers we tested

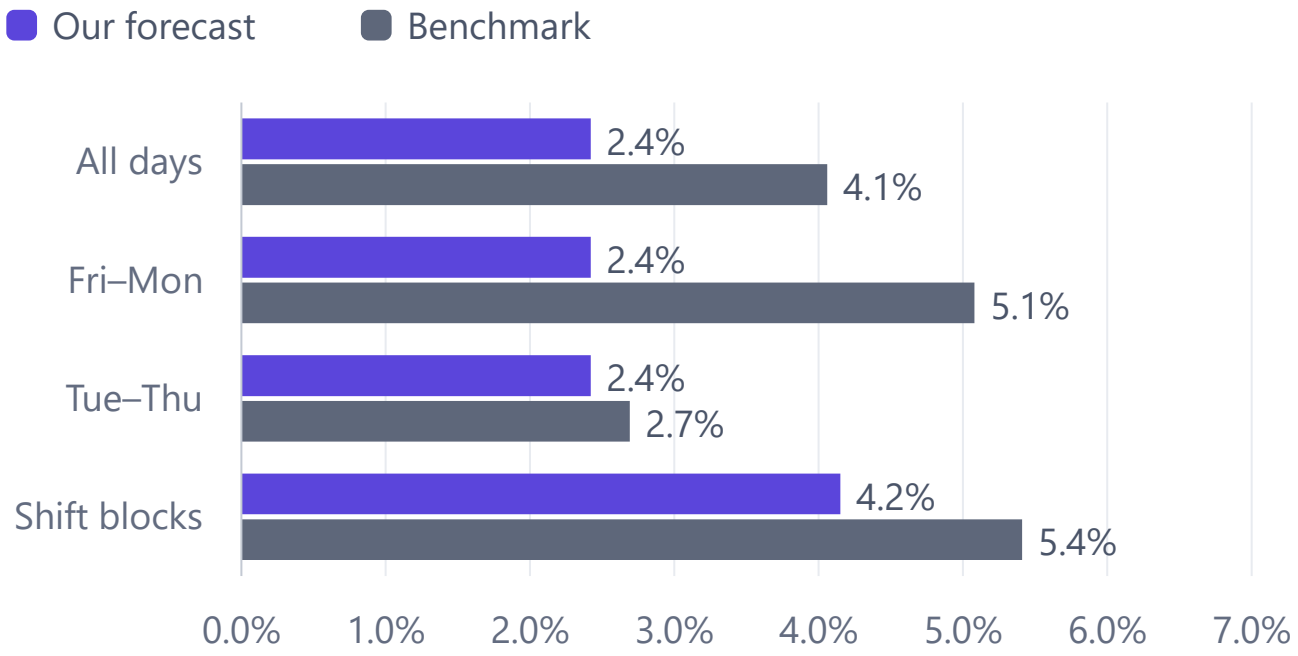
Driver	What we found	How it enters the plan
Weekday	R^2 0.977 on log daily demand with warehouse and weekday alone	In the model
Holidays	R^2 0.995 with holiday windows; MLK is the only analog	In the model, borrowed from MLK
Weekly level	R^2 0.999997; weekly levels ran 0.95 to 1.06, with no trend	Last 4 weeks' average level
Weather	Temperature adds nothing once weekday is known ($r = -0.006$)	Left out
Promotions	On every Fri–Sun and holiday, so inseparable ($r = -0.017$)	Calendar terms; PEAK's multipliers
Traffic	The data pack marks traffic_index an outcome: never a model input	Left out

Simulated data, not real Costco data

The forecast's edge on held-out weeks comes from removing holiday distortion

Mean absolute error on held-out weeks

Each week forecast from earlier weeks only. Lower is better.



Limits we state

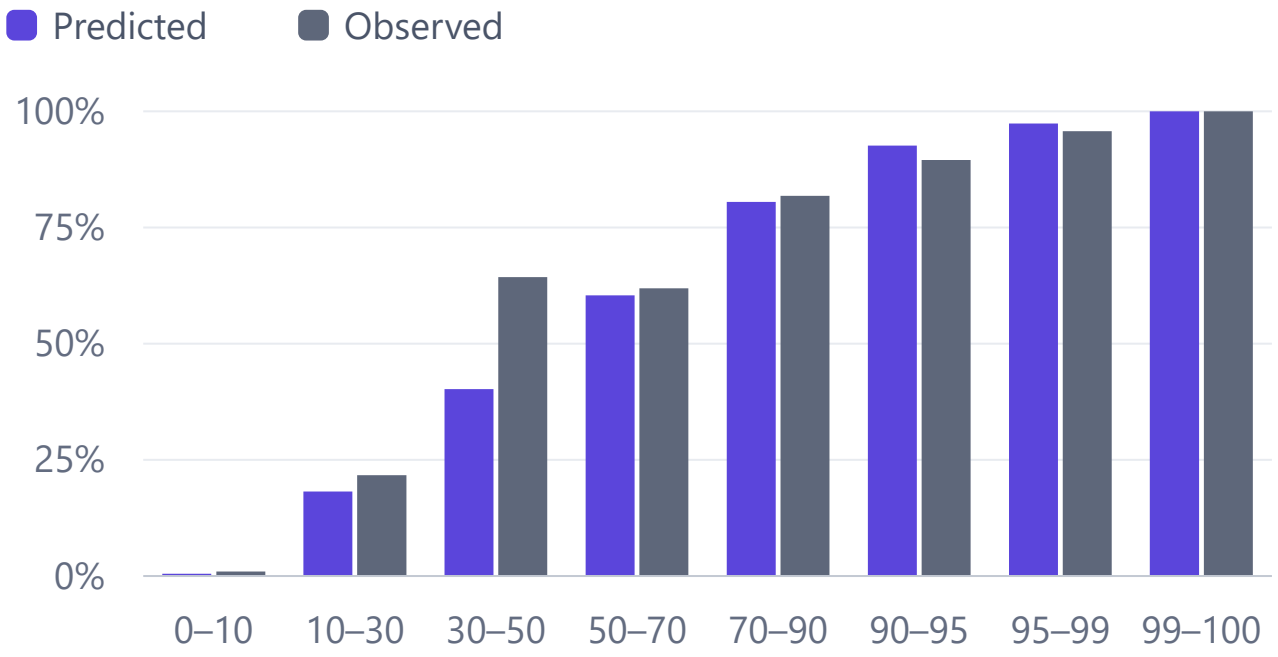
- The held-out weeks hold no holidays, and three folds is few. By week: 1.6% against 4.8%, 4.2% against 5.5%, 1.4% against 1.8%
- On normal days in the planning week, the two forecasts agree within 0.1%
- MLK weekend, the only holiday analog: 9.6% error against 10.2%, but worse for Pharmacy (11.9% against 7.0%)
- So we say the forecast removes holiday distortion, not that it beats the benchmark in the planning week

Simulated data, not real Costco data

A service curve fitted on history gives each block its chance of meeting target

Predicted against observed, out of sample

Blocks grouped by predicted chance of meeting target, in %



Utilization for a 95% chance

Department	W1	W2	W4
Front End	86.1%	83.7%	85.0%
Merchandising	85.7%	86.8%	84.5%
Pharmacy	85.5%	84.5%	84.7%

What each chance allows for

- Forecast error: 5.6%, 5.3% and 4.5% by department, more on holiday days
- Callouts by department and situation, with backfill while overtime lasts
- The curve's own scatter, with a smooth bend near the cliff (D-29)
- Calculated exactly, and checked against simulation. Brier score 0.024, against 0.177 for a flat rate

Simulated data, not real Costco data

Cost follows the case's rule, and Front End and Merchandising are solved together

Regular staff

First, at the loaded regular rate: $28.75 to $52.00 an hour

Cross-trained staff

Next, only people the partner department isn't using in that shift, at the regular rate

Overtime

Last, up to each block's limit: $37.95 to $68.64 an hour, about 32% more

How the search picks each shift's staffing

1. Highest total chance of meeting target, each block counted up to the 95% goal
2. Then the fewest hours short of need
3. Then the lowest cost

Pairs are tried together because the two departments share cross-trained staff.

One block's staff sources

- Warehouse 1, Front End, Sat Feb 14, Midday: 29 in the draft, 34 in the plan. Filled with 27 regular, 7 overtime.
- Cost: $3,647, which is $696 more than the draft.
- Across BASE, the plan uses 975 h of overtime and 95 h of cross-trained staff.

The plan adds 9 assumptions of our own to the case's, each with its basis

ID	Assumption	Basis	Whose
A-01	Cross-trained staff count only if the partner isn't using them	Our reading of the README	**Ours**
A-03	Plan for callouts: 5.2% call out, 49% of them backfilled	Six weeks of history	**Ours**
A-05	MLK weekend guides the Presidents' Day window	Same Friday-to-Monday shape	**Ours**
A-06	Valentine's Day adds 3% for Front End and Merchandising	No analog; tested for sensitivity	**Ours**
A-09	History's service curve holds in the planning week	The case says to estimate it	**Ours**
A-12	DISRUPT's illness brings January's flu-week callout rates	Both appeared together in January	**Ours**
A-13	The goal is a 95% chance of meeting target in each block	Service is judged first	**Ours**
A-14	Next week's level is the last 4 weeks' average	Best in the backtest	**Ours**
A-16	Some holiday-weekend PTO can still be limited	The data doesn't show approvals	**Ours**
	Labor standards, wage rates, shift hours and minimum staff	Data pack	Case
	Cost order: regular, then cross-trained, then overtime	Case, page 7, and README	Case
	Scenario demand multipliers and availability	Planning grid	Case

Against the draft, the plan raises expected service in every scenario and department

Scenario	Department	Hours draft	Hours plan	Cost draft	Cost plan	Cost change	Service draft	Service plan
BASE	**All**	10,344	**11,658**	$329.1K	**$377.5K**	**+14.7%**	68.5%	**95.6%**
	Front End	4,729	5,410	$139.2K	$163.3K	+17.3%	61.9%	**95.2%**
	Merchandising	4,846	5,404	$149.7K	$169.5K	+13.2%	58.5%	**95.0%**
	Pharmacy	770	844	$40.2K	$44.7K	+11.2%	87.7%	**96.6%**
PEAK	**All**	10,344	**11,923**	$329.1K	**$387.9K**	**+17.9%**	58.3%	**87.0%**
	Front End	4,729	5,494	$139.2K	$166.8K	+19.8%	49.1%	**79.4%**
	Merchandising	4,846	5,586	$149.7K	$176.4K	+17.8%	43.8%	**86.7%**
	Pharmacy	770	844	$40.2K	$44.7K	+11.2%	86.0%	**96.1%**
DISRUPT	**All**	10,207	**11,288**	$326.4K	**$366.6K**	**+12.3%**	66.6%	**89.5%**
	Front End	4,623	5,065	$137.0K	$152.7K	+11.4%	58.5%	**80.2%**
	Merchandising	4,834	5,392	$150.0K	$169.7K	+13.1%	58.3%	**94.5%**
	Pharmacy	750	832	$39.3K	$44.2K	+12.5%	85.6%	**94.5%**

DISRUPT's draft cost counts only staff who can be fielded: 36 people in 12 blocks can't be.

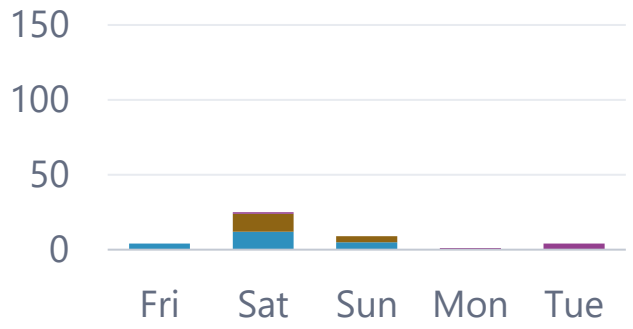
Simulated data, not real Costco data

Capacity runs out on the holiday weekend, most of all on Sat Feb 14

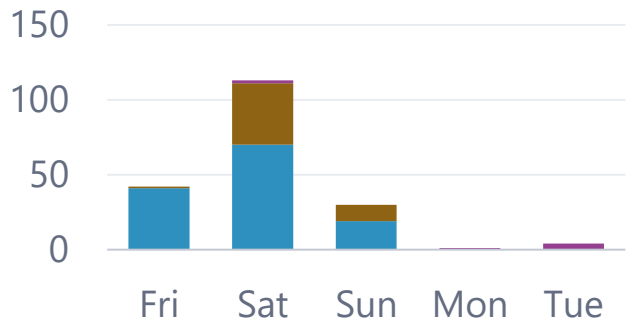
Front End Merchandising Pharmacy

People short of need in blocks below the goal, at full capacity

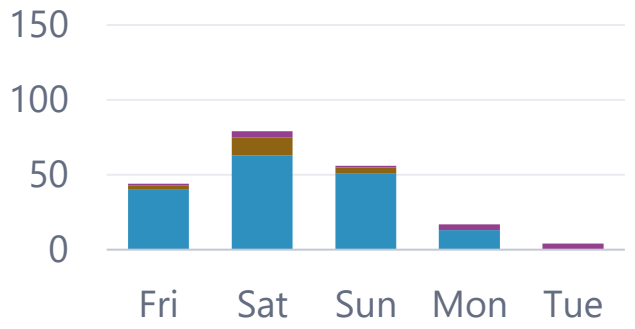
BASE: 43 people short



PEAK: 190 people short



DISRUPT: 200 people short



11 of 18

Front End and Merchandising blocks already at capacity on Sat Feb 14, BASE

4.2

fewer Front End regular staff per block than on recent weekends

83%

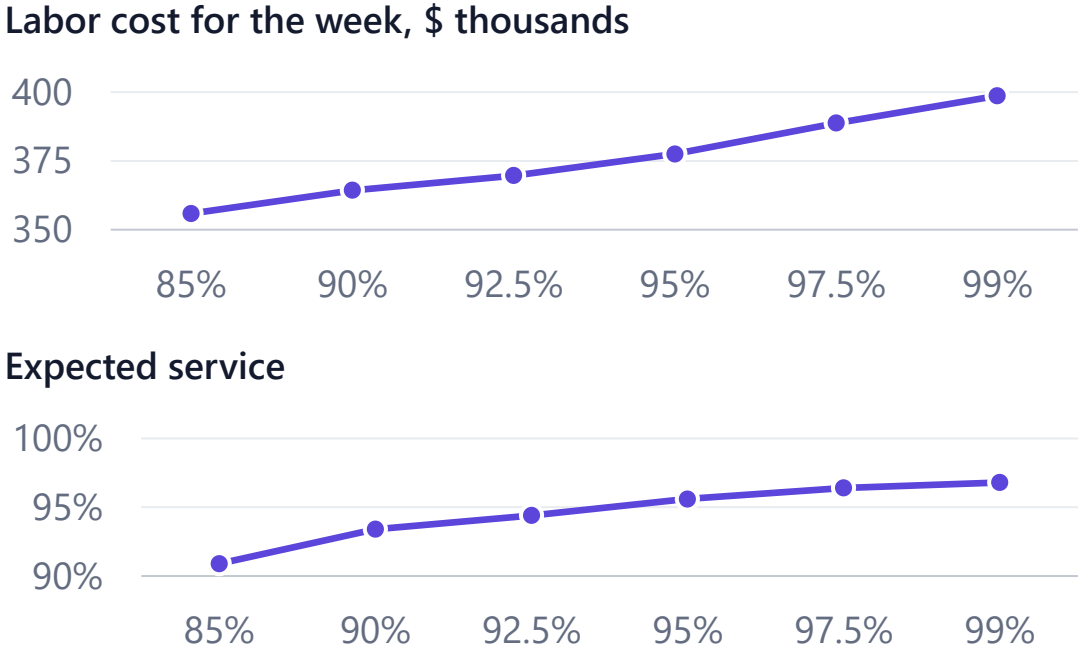
average chance in BASE's 25 short blocks; 11 of them sit at 90 to 95%

Simulated data, not real Costco data

Above a 95% goal each point of service costs more, and the callout allowance buys service

Service goal trade-off, BASE

Each point is the full plan rebuilt for that goal



What allowing for risk buys, BASE

Expected service when each risk occurs

Plan allows for	Cost	Both risks	Forecast only	Neither
Both (ours)	**$378K**	**96%**	**100%**	**100%**
Forecast error	$355K	90%	98%	100%
Neither	$328K	73%	86%	99%
Current draft	$329K	69%	77%	80%

The callout allowance costs $22.7K. Moving hours alone, at about the draft's cost, already beats the draft under every condition.

Simulated data, not real Costco data

Every check we could run passes: the file, the costs, the search and the risk model

Submission file

567 rows joined on keys, whole numbers, and all 27 closed blocks at 0. Our validator reports 0 problems.

Feasible everywhere

All 540 open blocks are within capacity and at or above minimum staff, in all three scenarios.

Cost rule

Rebuilt row by row from the README's rule; it matches every block and every total to the half cent.

Search

An independent brute force over every feasible pair picks the same staffing in every shift checked.

Risk math

The exact chance calculation agrees with a Monte Carlo simulation of the same risks.

Calibration

Out of sample, blocks predicted at 95 to 99% met target 95.7% of the time, against 97.4% predicted.

Explanations

A test checks every number quoted in all 567 block explanations against the plan.

Reproducible

Rerunning the pipeline rewrites every output byte for byte; the numbers were frozen before the deck.

The data is simulated, and a few of its quirks shaped the method

Simulated data

A competition dataset, not Costco data. Wage rates are mock values. Every figure in this deck comes from it.

DISRUPT cuts more

36 blocks lose staff, including partner cross-trained staff the docs don't mention, so we read availability per scenario from the grid.

Promotions follow the calendar

Promotions run on every Friday to Sunday and holiday, so their effect can't be separated from the weekday and holiday terms.

Traffic is an outcome

The data pack labels traffic_index an output, never a model input, because it would leak the answer. We left it out.

Holiday labels differ

History names MLK weekend and the planning week names Presidents' Day, so we mapped both to shared holiday roles.

One draft for all scenarios

The draft schedules the same 2,390 employee shifts in all three scenarios, whatever the conditions.

The submission file holds all 567 blocks, joined on keys, with closed blocks at zero

A sample of rows, in the template's own order

Scenario	Warehouse	Department	Date	Shift	Draft	Submitted
BASE	1	Front End	Feb 14	Evening	24	27
BASE	1	Front End	Feb 14	Midday	29	34
BASE	1	Front End	Feb 14	Opening	23	27
BASE	1	Pharmacy	Feb 14	Evening	0	0
BASE	1	Pharmacy	Feb 14	Midday	2	3
BASE	1	Pharmacy	Feb 14	Opening	3	4
DISRUPT	2	Front End	Feb 14	Evening	24	20
DISRUPT	2	Front End	Feb 14	Midday	31	29
DISRUPT	2	Front End	Feb 14	Opening	27	21

Closed blocks show 0. "Submitted" fills the template's submitted_employees column.

What the file holds

- 567 rows: 3 scenarios of 189 blocks
- Joined on scenario, warehouse, department, date and shift, never by row order
- Total employees per block, as the template asks, with no split by staff type
- 27 closed Pharmacy blocks set to 0
- Employee shifts, draft to plan: BASE 2,390 to 2,710; PEAK 2,390 to 2,766; DISRUPT 2,390 to 2,605